

BCS703: Cryptography and Network Security (Module 1)

Question 1:

Obtain the ciphertext for a given plaintext using the Hill Cipher and a given key matrix. Show all matrix operations.

$$K = \begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$$

Answer:

1. Introduction

The **Hill Cipher** is a multiletter substitution cipher developed by **Lester Hill in 1929**. It uses **matrix arithmetic modulo 26**. Each letter is assigned a numerical value:

$$a = 0, b = 1, c = 2, \dots, z = 25$$

For a 3×3 Hill Cipher, three plaintext letters are processed at a time.

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2. Hill Cipher Encryption Formula

The encryption operation is represented as:

$$C = PK \pmod{26}$$

where:

- P = plaintext row vector
- C = ciphertext row vector
- K = encryption key matrix
- All matrix operations are performed modulo 26.

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For a 3×3 key matrix,

$$K = \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix}$$

and

$$P = (p_1 \ p_2 \ p_3)$$

then

$$C = (c_1 \ c_2 \ c_3)$$

where



$$c_1 = (k_{11}p_1 + k_{21}p_2 + k_{31}p_3) \pmod{26}$$

$$c_2 = (k_{12}p_1 + k_{22}p_2 + k_{32}p_3) \pmod{26}$$

$$c_3 = (k_{13}p_1 + k_{23}p_2 + k_{33}p_3) \pmod{26}$$

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3. Given Example

The uploaded notes give the following example:

Plaintext: paymoremoney

Key matrix:

$$K = \begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$$

The notes use the first three plaintext letters **pay**, represented by:

$$P = (15 \ 0 \ 24)$$

because

$$p = 15, \quad a = 0, \quad y = 24$$

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4. Matrix Multiplication

Using

$$C = PK \pmod{26}$$

we get

$$C = (15 \ 0 \ 24) \begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$$

Now perform the multiplication:

$$\begin{aligned} C &= (15(17) + 0(21) + 24(2), \\ &\quad 15(17) + 0(18) + 24(2), \\ &\quad 15(5) + 0(21) + 24(19)) \end{aligned}$$

Therefore,

$$\begin{aligned} C &= (255 + 48, 255 + 48, 75 + 456) \\ C &= (303, 303, 531) \end{aligned}$$

Now take each value modulo 26:

$$303 \bmod 26 = 17$$

$$303 \bmod 26 = 17$$

$$531 \bmod 26 = 11$$

Hence,

$$\boxed{C = (17 \ 17 \ 11)}$$

Using $a = 0, b = 1, \dots, z = 25$:

$$17 = R, \quad 17 = R, \quad 11 = L$$

Therefore,

$$\boxed{\text{PAY} \rightarrow \text{RRL}}$$

5. Encrypting the Complete Plaintext

The plaintext is:

PAY MORE MONEY

Ignoring spaces:

PAYMOREMONEY

Divide it into groups of three:

Plaintext group	Numerical vector
PAY	(15, 0, 24)
MOR	(12, 14, 17)
E M O	(4, 12, 14)
NEY	(13, 4, 24)

Each group is multiplied by the same key matrix:

$$C_i = P_i K \pmod{26}$$

For example, the first group gives:

$$\begin{aligned}(15 \ 0 \ 24)K &= (303 \ 303 \ 531) \pmod{26} \\ &= (17 \ 17 \ 11) \\ &= \text{RRL}\end{aligned}$$

Repeating this operation for all plaintext groups produces the ciphertext:

RRLMWBKASPDH

6. General Procedure for Hill Cipher Encryption

Step 1: Assign numerical values to the plaintext letters:

$$a = 0, b = 1, \dots, z = 25$$

Step 2: Divide the plaintext into groups of m letters, where m is the dimension of the key matrix.

Step 3: Convert every group into a plaintext row vector P .

Step 4: Multiply the plaintext vector with the key matrix:

$$C = PK$$

Step 5: Perform all calculations modulo 26:

$$C = PK \pmod{26}$$

Step 6: Convert the resulting numerical values back into letters.

Step 7: Concatenate the resulting ciphertext groups to obtain the final ciphertext.

The notes describe the Hill algorithm as taking m successive plaintext letters and substituting them with m ciphertext letters using m linear equations. [BCS703-module-1-textbook\(1\)](#)

7. Final Answer

For the given example:

Plaintext = PAYMOREMONEY

$$K = \begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$$

Using

$$C = PK \pmod{26}$$

the resulting ciphertext is:

RRLMWBKASPDH

Question 2:

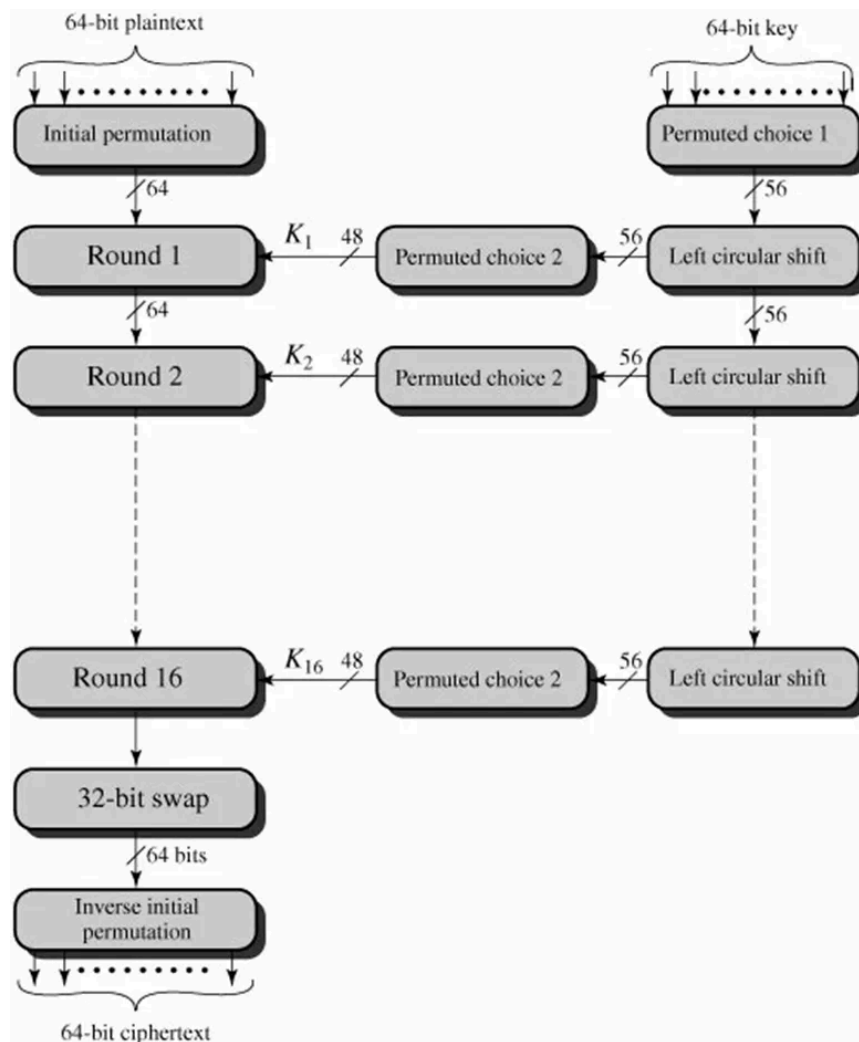
Explain the Data Encryption Standard (DES) algorithm with a neat block diagram. Explain DES key generation and the 16-round structure.

Answer:

Data Encryption Standard (DES)

The **Data Encryption Standard (DES)** is a symmetric-key block cipher based on the **Feistel structure**. It operates on a **64-bit plaintext block** using a **64-bit key input**, of which only **56 bits are actually used**; the remaining 8 bits may serve as parity bits. DES performs **16 rounds** of processing using different 48-bit round subkeys.

DES Encryption : Block Diagram



2. DES Encryption Process

DES encryption consists of **three major phases**:

1. Initial permutation
2. Sixteen Feistel rounds
3. Final permutation

The 64-bit plaintext first passes through the **Initial Permutation (IP)**. The resulting 64-bit block is divided into two 32-bit halves:

$$IP(P) = L_0 || R_0$$

The two halves then undergo 16 rounds of processing. Finally, the halves are swapped and passed through the **Inverse Initial Permutation IP^{-1}** to obtain the 64-bit ciphertext.

3. DES 16-Round Structure

Each round uses:

- L_{i-1} — previous left half
- R_{i-1} — previous right half
- K_i — round key

The Feistel equations are:

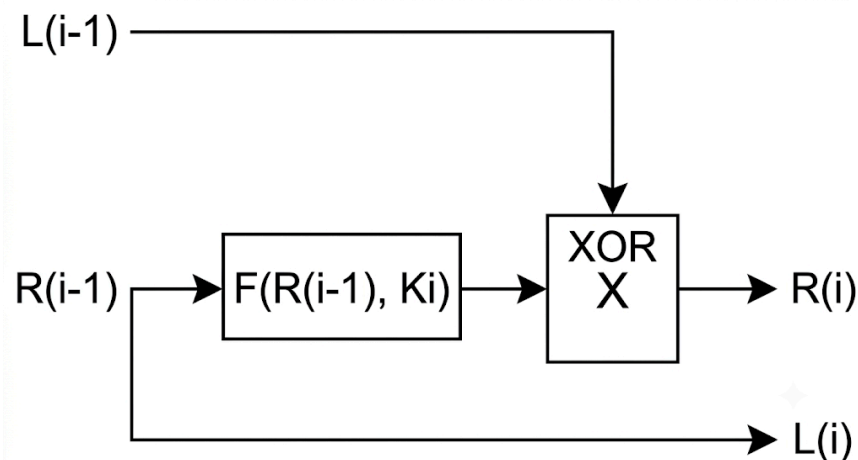
$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

where F is the DES round function.

The textbook describes each round as applying a function F to the right half, XORing its output with the left half, and then interchanging the two halves. [BCS703-module-1-textbook\(1\)](#)

One DES Round



Thus:

$$L_i = R_{i-1}$$

and

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

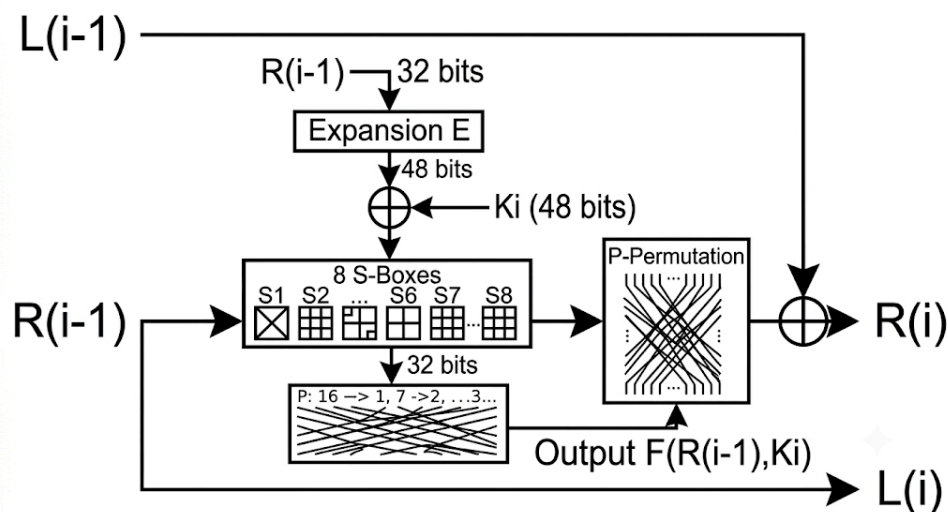
This structure is repeated for 16 rounds.

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4. DES Round Function FF

The DES round function operates on the **32-bit right half** and the **48-bit round key**.

The standard structure is:



The **S-boxes** provide the nonlinear substitution component and are central to the confusion provided by DES. The eight substitution tables/S-boxes is part of each DES iteration.

5. DES Key Generation

DES receives a **64-bit key**, but only **56 bits** are used. The other 8 bits are parity bits or otherwise unused.

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Step 1: Permuted Choice 1 (PC-1)

The 64-bit key is subjected to **Permuted Choice 1**.

$$64\text{-bit key} \xrightarrow{PC^{-1}} 56\text{-bit key}$$

The 56 bits are divided into two 28-bit halves:

$$C_0 = 28 \text{ bits}$$

$$D_0 = 28 \text{ bits}$$

Step 2: Left Circular Shift

For each round, C_{i-1} and D_{i-1} undergo a **left circular shift**.

$$C_{i-1} \rightarrow C_i$$

$$D_{i-1} \rightarrow D_i$$

The number of shifts depends on the round.

Step 3: Permuted Choice 2 (PC-2)

The shifted C_i and D_i are combined:

$$C_i || D_i = 56 \text{ bits}$$

This 56-bit result is subjected to **Permuted Choice 2 (PC-2)**:

$$56 \text{ bits} \xrightarrow{PC-2} 48 \text{ bits}$$

This produces the round key:

$$K_i = 48 \text{ bits}$$

Therefore:

$$K_1, K_2, K_3, \dots, K_{16}$$

are generated for the 16 rounds.

8. Final Swap and Inverse Permutation

After Round 16, the two 32-bit halves are swapped:

$$L_{16} || R_{16} \longrightarrow R_{16} || L_{16}$$

This is then passed through the inverse initial permutation:

$$C = IP^{-1}(R_{16} || L_{16})$$

The result is the final **64-bit ciphertext**. [BCS703-module-1-textbook\(1\)](#)

9. DES Decryption

One important advantage of the Feistel structure is that the same basic algorithm can be used for decryption.

The only major difference is the order of the subkeys:

$$K_{16}, K_{15}, \dots, K_2, K_1$$

are used instead of:

$$K_1, K_2, \dots, K_{15}, K_{16}$$

The textbook explicitly states that DES decryption uses the same algorithm with the subkeys applied in reverse order. [BCS703-module-1-textbook\(1\)](#)

Question 3:

State the rules used for encryption in the Playfair cipher and encrypt a given message using a specified keyword.

Answer:

Playfair Cipher : Encryption Rules and Example

The **Playfair cipher** is a digraph substitution cipher. Unlike a simple substitution cipher, it encrypts **two plaintext letters at a time**. It uses a **5×5 matrix** constructed from a keyword, with **I and J treated as one letter**.

Lets take example as **MONARCHY** and demonstrate the encryption rules using the plaintext **BALLOON**.

1. Constructing the Playfair Matrix

Keyword: MONARCHY

First write the keyword, removing repeated letters:

MONARCHY

Then fill the remaining alphabetic letters in order, excluding letters already used and treating **I/J** as **one letter**.

The resulting matrix is:  BCS703-module-1-textbook(1)

<i>M</i>	<i>O</i>	<i>N</i>	<i>A</i>	<i>R</i>
<i>C</i>	<i>H</i>	<i>Y</i>	<i>B</i>	<i>D</i>
<i>E</i>	<i>F</i>	<i>G</i>	<i>I/J</i>	<i>K</i>
<i>L</i>	<i>P</i>	<i>Q</i>	<i>S</i>	<i>T</i>
<i>U</i>	<i>V</i>	<i>W</i>	<i>X</i>	<i>Z</i>

2. Rules Used for Encryption

The plaintext is divided into **pairs of letters (digraphs)**. Each pair is encrypted according to one of the following four rules.  BCS703-module-1-textbook(1)

Rule 1: Repeated Letters

If both letters in a pair are the same, insert a **filler letter**, usually **X**, between them.

For example:

BALLOON

cannot be divided directly as:

BA LL OO N

because **LL** and **00** contain repeated letters.

Therefore, insert **X**:

BA LX LO ON

The textbook gives the same treatment for **BALLOON**, representing it as:

BA LX LO ON

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Rule 2: Same Row

If the two letters are in the **same row**, replace each letter with the letter immediately to its **right**.

The row is considered circular, so the letter after the last element is the first element.

For example, in the matrix:

L P Q S T

If the pair is:

LP

then:

$L \rightarrow P$

$P \rightarrow Q$

Therefore:

$LP \rightarrow PQ$

Rule 3: Same Column

If the two letters are in the **same column**, replace each letter with the letter immediately **below** it.

The column is circular, so the bottom element is followed by the top element.

For example:

B

and

I/J

are in the same column:

A
B
I/J
S
X

Thus:

$B \rightarrow I/J$

and

$A \rightarrow B$

The textbook gives $MU \rightarrow CM$ as an example of the same-column rule. [BCS703-module-1-textbook\(1\)](#)

Rule 4: Different Row and Different Column

If the two letters are **not in the same row or column**, form the rectangle whose corners contain the two letters.

Rule 4: Different Row and Different Column

If the two letters are **not in the same row or column**, form the rectangle whose corners contain the two letters.

Each letter is replaced by the letter in:

its own row + the other letter's column

For example, consider **HS**.

- H = row 2, column 2
- S = row 4, column 4

Therefore:

$$H \rightarrow B$$

because B is in H's row and S's column.

Similarly:

$$S \rightarrow P$$

because P is in S's row and H's column.

Hence:

$$HS \rightarrow BP$$

This is exactly the rectangle rule described in the textbook.

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3. Encryption of BALLOON

Step 1: Divide plaintext into pairs

Plaintext:

BALLOON

Because repeated letters cannot occur in the same pair:

BA LX LO ON

Step 2: Encrypt **BA**

Positions:

- B → row 2, column 4
- A → row 1, column 4

They are in the **same column**.

Apply the same-column rule:

$B \rightarrow I/J$

$A \rightarrow B$

Therefore:

BA → IB

(or **JB** depending on the I/J convention).

Step 3: Encrypt **LX**

Positions:

- $L \rightarrow$ row 4, column 1
- $X \rightarrow$ row 5, column 4

Different row and different column.

Use the rectangle rule:

$$L \rightarrow X$$

$$X \rightarrow L$$

Therefore:

$$\boxed{LX \rightarrow XL}$$

Step 4: Encrypt **LO**

Positions:

- $L \rightarrow$ row 4, column 1
- $O \rightarrow$ row 1, column 2

Different row and column.

Rectangle rule:

$$L \rightarrow P$$

$$O \rightarrow M$$

Therefore:

$$\boxed{LO \rightarrow PM}$$

Step 5: Encrypt ON

Positions:

- O → row 1, column 2
- N → row 1, column 3

They are in the **same row**.

Move each one position to the right:

$$O \rightarrow N$$

$$N \rightarrow A$$

Therefore:

$$\boxed{ON \rightarrow NA}$$

4. Final Ciphertext

Combining all the encrypted digraphs:

$$BA \rightarrow IB$$

$$LX \rightarrow XL$$

$$LO \rightarrow PM$$

$$ON \rightarrow NA$$

Therefore:

$$\boxed{BALLOON \rightarrow IBXLPMNA}$$

The choice of **I** versus **J** in the first pair depends on the convention used for the combined I/J cell.

Summary of Playfair Rules

Condition	Encryption rule
Repeated letters in a pair	Insert X between them
Same row	Replace each letter by the letter to its right

Condition	Encryption rule
Same column	Replace each letter by the letter below it
Different row and column	Use the rectangle rule
Matrix	5×5
I/J	Treated as one letter
Processing	Two letters at a time

Question 4:

Explain the model for network security with a neat diagram and describe the roles of sender, receiver, opponent, security transformation, secret information and trusted third party.

Answer:

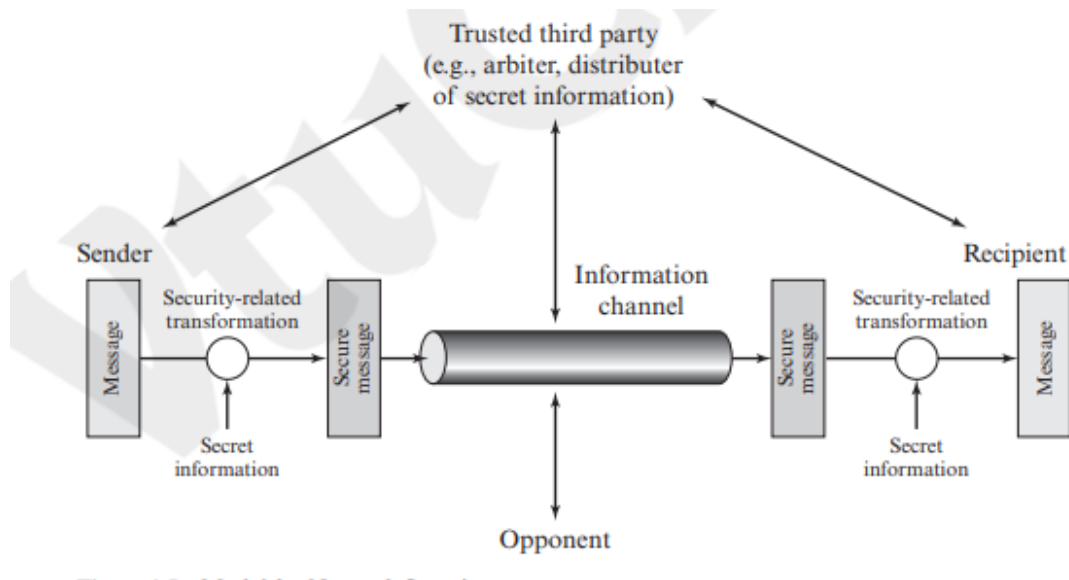
Model for Network Security

1. Introduction

A **model for network security** describes how a message is securely transferred from a **sender** to a **receiver** through an information channel in the presence of an **opponent** who may threaten the security of the communication.

The model uses a **security-related transformation** and **secret information** to protect the transmitted message. A **trusted third party** may also be required for distributing secret information or resolving disputes about message authenticity.

2. Neat Block Diagram



The sender and recipient communicating through an information channel, with security-related transformations and secret information at the two ends. The opponent attempts to threaten the protected communication, while a trusted third party may distribute secret information or act as an arbiter.

3. Components of the Model

1. Sender

The **sender** is the party that originates the message.

The sender applies a **security-related transformation** to the message before transmitting it through the information channel.

For example, encryption can transform the original message into an unreadable form so that an opponent cannot understand it.

2. Receiver

The **receiver** is the intended destination of the message.

The receiver applies the appropriate security-related transformation using the required secret information to obtain or verify the original message.

Thus, the receiver must be able to interpret the protected message correctly.

3. Opponent

The **opponent** is an unauthorized party who may attempt to attack or interfere with the communication.

The purpose of introducing security into the model is to protect the information transmission from an opponent who may threaten properties such as:

- **Confidentiality**
- **Authenticity**

The opponent should not be able to defeat the purpose of the security transformation.

4. Security Transformation

A **security-related transformation** is applied to the information to provide the required security service.

Examples include:

- **Encryption** — scrambles the message so that it is unreadable to the opponent.
- **Addition of a code based on message contents** — can be used to verify the identity of the sender.

Thus, the transformation protects the message during transmission.

5. Secret Information

Secret information is information shared by the communicating principals and, ideally, kept unknown to the opponent.

An example is an **encryption key**.

The key is used together with the security transformation to:

scramble the message before transmission

and

unscramble it at reception

Therefore, protecting the secret information is essential for secure communication.

6. Trusted Third Party

A **trusted third party** is an entity that may be required to support secure communication between the sender and receiver.

It can perform functions such as:

1. **Distributing secret information** to the communicating parties while keeping it away from the opponent.
2. **Arbitrating disputes** between the two parties concerning the authenticity of a message transmission.

Examples mentioned in the model include an **arbiter** or **distributor of secret information**.

4. Working of the Model

The operation can be summarized as follows:

Step 1: The sender generates a message.

Step 2: A security-related transformation is applied to the message using secret information.

Step 3: The resulting **secure message** is transmitted through the information channel.

Step 4: The opponent may attempt to attack or interfere with the transmission.

Step 5: The receiver applies the appropriate security-related transformation using the required secret information.

Step 6: A trusted third party may assist in distributing the secret information or resolving authenticity disputes.

5. Basic Tasks in Designing a Security Service

The model leads to **four basic tasks**:

1. **Design a security algorithm** for performing the security-related transformation such that an opponent cannot defeat its purpose.
 2. **Generate the secret information** required by the algorithm.
 3. **Develop methods for distributing and sharing** the secret information.
 4. **Specify a protocol** used by the two communicating parties that combines the security algorithm and secret information to provide the required security service.
-

Question 5:

Describe the One-Time Pad and simple XOR encryption techniques with examples and discuss their difficulties/practical limitations.

Answer:

One-Time Pad and Simple XOR Encryption

1. Introduction

The **Vernam cipher** is a substitution technique that works on **binary data (bits)** rather than letters. Its basic operation is the **bitwise XOR** of the plaintext and the key. The One-Time Pad is an improvement of the Vernam cipher in which the key is **truly random, as long as the message, used only once, and then discarded.**

2. Simple XOR Encryption

In XOR encryption, each plaintext bit is combined with the corresponding key bit using the **exclusive-OR (XOR)** operation.

The encryption equation is:

$$c_i = p_i \oplus k_i$$

where:

- $p_i = i^{th}$ plaintext bit
- $k_i = i^{th}$ key bit
- $c_i = i^{th}$ ciphertext bit
- $\oplus$ = XOR operation

The ciphertext is therefore generated by performing a **bitwise XOR** between plaintext and key.

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XOR Truth Table

P	K	$P \oplus K$
0	0	0
0	1	1
1	0	1
1	1	0

3. Example of XOR Encryption

Consider:

$$P = 10110010$$

and

$$K = 01101100$$

Perform bitwise XOR:

```
Plaintext : 10110010
Key       : 01101100
          -----
Ciphertext: 11011110
```



Therefore,

$$C = 11011110$$

Decryption

A major property of XOR is:

$$P = C \oplus K$$

Hence:

```
Ciphertext: 11011110
Key       : 01101100
          -----
Plaintext : 10110010
```



Thus the original plaintext is recovered.

The notes explicitly state that because of the properties of XOR, **the same XOR operation is used for decryption**:

$$p_i = c_i \oplus k_i$$

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4. Vernam Cipher and Its Limitation

Gilbert Vernam introduced a system in which a keyword was made as long as the plaintext and had no statistical relationship to it. BCS703-module-1-textbook(1)

Vernam originally proposed a **running loop of tape**, causing the key to eventually repeat. Thus, although the key could be very long, it was ultimately a **repeating keyword**.

This creates a weakness: with sufficient ciphertext and knowledge of **known or probable plaintext sequences**, the system can be broken. BCS703-module-1-textbook(1)

So:

Repeating XOR key $\neq$ One-Time Pad

This distinction is important in an exam.

5. One-Time Pad

The **One-Time Pad (OTP)** was proposed by **Joseph Mauborgne** as an improvement to the Vernam cipher.

It uses a:

- **Random key**
- **Key that is as long as the message**
- **Key that is used for only one message**
- **Key that is discarded after use**
- **New key of equal length** for every new message

 BCS703-module-1-textbook(1)

The encryption operation can be represented as:

$$C = P \oplus K$$

and decryption as:

$$P = C \oplus K$$

6. Example of One-Time Pad

Consider the plaintext:

$$P = 10110010$$

Let the randomly generated key be:

$$K = 01101100$$

Encryption

$$C = P \oplus K$$

$$\begin{array}{rcl} P & : & 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 0 \\ K & : & 0 \ 1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \\ \hline C & : & 1 \ 1 \ 0 \ 1 \ 1 \ 1 \ 1 \ 0 \end{array}$$

Therefore:

$$C = 11011110$$

Decryption

The receiver uses the **same key**:

$$P = C \oplus K$$

$$\begin{array}{rcl} C & : & 1 \ 1 \ 0 \ 1 \ 1 \ 1 \ 1 \ 0 \\ K & : & 0 \ 1 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \\ \hline P & : & 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 0 \end{array}$$

Thus:

$$P = 10110010$$

The important point is that the key must be **truly random and never reused**. The notes state that the security of the OTP is entirely due to the randomness of its key. [BCS703-module-1-textbook\(1\)](#)

7. Why One-Time Pad Is Unbreakable

The One-Time Pad produces ciphertext that has **no statistical relationship with the plaintext** when the key is truly random.

For example, the same ciphertext can correspond to different plaintexts when different keys are used.

We can demonstrate this with a ciphertext for which two different random keys produce two different meaningful plaintexts. Given any plaintext of the same length as the ciphertext, there exists a corresponding key that can produce it. Therefore, examining all possible keys does not reveal which plaintext is the correct one.

Hence:

One-Time Pad provides perfect secrecy

8. Practical Difficulties / Limitations

Although OTP provides complete security in theory, it has major practical problems.

1. Generation of Random Keys

A large amount of **truly random key material** is required.

For heavily used systems, millions of random characters may have to be generated regularly. Producing truly random material in such quantities is difficult.

2. Key Distribution

This is an even bigger problem.

For **every message**, both sender and receiver must possess a key that is:

equal in length to the message

Therefore, a huge amount of key material has to be securely distributed and protected.

3. Key Protection

The key must remain secret. If an opponent obtains the key, the security of the corresponding communication is lost.

The symmetric encryption model emphasizes that the sender and receiver must obtain the secret key securely and keep it secure.

4. Limited Practical Utility

Because of the difficulty of generating, distributing, and protecting such large quantities of random keys, OTP has **limited practical utility**.

The notes state that it is mainly useful for **low-bandwidth channels requiring very high security**.

9. Simple XOR vs One-Time Pad

Feature	Simple XOR / Vernam	One-Time Pad
Operation	XOR	XOR
Key	May be a repeating key	Truly random
Key length	Can be shorter and repeated	Equal to message length
Key reuse	Repetition possible	Never reused
Security	Can be broken with sufficient information	Unbreakable in theory
Main weakness	Repeated key creates patterns	Key generation/distribution
Security property	Depends on key construction	Perfect secrecy

The crucial difference is **not the XOR operation**. Both use XOR. The security difference comes from the **construction and use of the key**. The repeating Vernam key can be broken, while the random, nonrepeating OTP provides complete security

Question 6:

Explain Steganography and discuss its advantages and disadvantages.

Answer:

Steganography

1. Definition

Steganography is the technique of **hiding the existence of a message** by embedding the secret information inside another ordinary-looking medium, called the **cover medium**.

Unlike encryption, which makes a message unreadable, steganography attempts to make the message **undetectable**.

For example, a secret message can be hidden inside an image so that the image appears normal to an observer.

2. How Steganography Works

The general procedure is:

Step 1: Select a cover medium

An ordinary image, audio file, video, or text is selected.

Step 2: Select the secret message

The information to be hidden is prepared.

Step 3: Embed the message

A steganographic algorithm modifies the cover medium in such a way that the secret message is embedded while keeping the changes difficult to notice.

Step 4: Generate the stego medium

The resulting file looks like the original cover medium.

Step 5: Extract the message

The intended receiver uses the appropriate extraction procedure to recover the hidden information.

3. Example

Suppose we want to hide:

| **MEET**

inside an ordinary image.

The image acts as the **cover medium**.

The characters of **MEET** are converted into binary data and embedded into suitable bits of the image.

Original Image

+

Secret Message



Steganographic Embedding



Stego Image

To a normal viewer, the stego image may look almost identical to the original image, while the receiver can extract **MEET**.

4. Common Forms of Steganography

Steganography can be applied to different types of media:

Cover medium	Secret information hidden in
Image	Pixel data
Audio	Audio samples
Video	Video frames
Text	Formatting/character patterns
Network traffic	Communication characteristics

A common digital approach is to modify the **least significant bits (LSBs)** of data. Small changes in these bits generally have little perceptible effect on the cover medium.

6. Advantages of Steganography

1. Hides the existence of communication

Its major advantage is that it does not merely hide the contents; it attempts to hide the **fact that secret communication is taking place**.

2. Can provide an additional security layer

Steganography can be combined with encryption:

Plaintext → Encryption → Steganography

Even if the hidden data is discovered, the attacker may still need to decrypt it.

3. Maintains the appearance of the original medium

A properly designed stego medium can appear normal to an observer.

4. Can use different types of media

Images, audio, video, and other digital data can serve as carriers.

7. Disadvantages of Steganography

1. Limited capacity

Only a limited amount of secret information can normally be hidden without significantly altering the cover medium.

Trying to hide too much data can make the modifications detectable.

2. Vulnerable to detection

Steganalysis techniques can analyze files and identify statistical abnormalities caused by hidden information.

Therefore, steganography is **not automatically secure simply because the message cannot be seen directly.**

3. Modification of the cover can destroy the hidden message

Operations such as:

- Compression
- Resizing
- Filtering
- Format conversion
- Noise addition

may damage or completely destroy the embedded information.

4. Requires a suitable cover medium

The effectiveness of the technique depends on the properties and size of the carrier.

5. Does not necessarily provide confidentiality

This is an important distinction:

Steganography hides existence

whereas

Encryption hides meaning

If an attacker successfully detects and extracts the hidden message, the message may be readable unless it was also encrypted.

Question 7:

Explain monoalphabetic substitution ciphers and discuss their vulnerability to frequency analysis.

Answer:

Monoalphabetic Substitution Cipher

1. Definition

A **monoalphabetic substitution cipher** is a substitution cipher in which **each plaintext letter is replaced by a fixed corresponding ciphertext letter throughout the entire message.**

Thus, a single cipher alphabet is used for the complete message.

For example:

```
Plain :  ABCDEFGHIJKLMNOPQRSTUVWXYZ  
Cipher:  QWERTYUIOPASDFGHJKLZXCVBNM
```

If:

A → Q

then every occurrence of **A** in the plaintext is replaced by **Q**.

The key is essentially a **permutation of the 26-letter alphabet**, giving:

26!

possible substitution alphabets. This is greater than 4×10^{26} which makes a simple brute-force attack impractical.

2. Encryption

Let:

- P = plaintext
- C = ciphertext
- K = substitution key

Encryption can be represented as:

$$C = E(K, P)$$

Each plaintext character is replaced by its corresponding character in the cipher alphabet.

Example

Suppose the substitution is:

Plain	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
Cipher	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>

Then:

BAD

becomes:

$B \rightarrow E$

$A \rightarrow D$

$D \rightarrow G$

Therefore:

$$\text{BAD} \rightarrow \text{EDG}$$

The important characteristic is that **the same plaintext letter always maps to the same ciphertext letter**.

3. Why Monoalphabetic Cipher Appears Secure

There are:

$$26!$$

possible permutations of the alphabet.

Since:

$$26! > 4 \times 10^{26}$$

trying every possible key by brute force would be computationally impractical for a classical cipher.

BCS703-module-1-textbook(1)

However, this does **not** make the cipher secure.

The weakness is that the cipher preserves the **statistical structure of the plaintext language**.

4. Vulnerability to Frequency Analysis

Basic Idea

In normal English text, some letters occur much more frequently than others.

For example, letters such as:

E, T, A, O, I

occur frequently, while letters such as:

Q, X, Z

occur relatively rarely.

A monoalphabetic substitution does not change the frequency pattern; it only changes the identity of the letters.

For example, if **E** is replaced by **X**, then **every E becomes X**.

Therefore, the most frequent ciphertext character may correspond to the most frequent plaintext character, such as **E**.

If the plaintext is ordinary English, the cryptanalyst can exploit the **regularities of the language**.

5. Frequency Analysis Procedure

A cryptanalyst can attack the cipher approximately as follows:

Step 1: Count ciphertext letter frequencies

Determine how often each ciphertext character occurs.

Example:

Ciphertext frequency:

X → 13%

Q → 10%

M → 8%

...

Step 2: Compare with English frequencies

Compare the observed frequencies with the expected frequencies of English letters.

If **X** is the most frequent ciphertext character, it may represent **E**.

Thus, initially:

X → EX

Step 3: Examine common patterns

The attacker can examine repeated patterns and common words.

For example, a one-letter word in English is likely to be:

A or I

Common three-letter patterns may correspond to words such as:

THE, AND, FOR

Step 4: Test the substitutions

The cryptanalyst substitutes the suspected plaintext letters into the ciphertext.

If the resulting text begins to form meaningful English words, additional mappings can be inferred.

Step 5: Refine the mapping

The substitutions are repeatedly adjusted until a consistent plaintext emerges.

6. Example of Frequency Analysis

Suppose the ciphertext contains:

```
WZXL XLMW WZXL ...
```

Assume **X** is the most frequent character.

The attacker may hypothesize:

X=E

If a frequently occurring pattern is:

XLMW

then it has the pattern:

E _ _ _

By combining:

- Letter frequency
- Repeated-letter patterns
- Common word patterns
- Knowledge of English

the cryptanalyst can gradually determine the substitution alphabet.

This is why simply having a huge keyspace does not make monoalphabetic substitution secure.

9. Advantages

1. Large keyspace

There are:

26!

possible substitution alphabets.

2. Simple implementation

The encryption and decryption operations are straightforward.

3. Better than Caesar cipher

Unlike Caesar cipher, which has only a small number of possible shifts, arbitrary substitution provides a dramatically larger keyspace.

10. Disadvantages

1. Vulnerable to frequency analysis

The primary weakness is that the statistical characteristics of the plaintext survive in the ciphertext.

2. Patterns are preserved

Repeated plaintext letters produce repeated ciphertext letters.

3. Word patterns can reveal plaintext

Repeated words and common letter combinations provide clues.

4. Not suitable for modern secure communication

Despite its large keyspace, the cipher can be attacked using statistical and linguistic analysis.

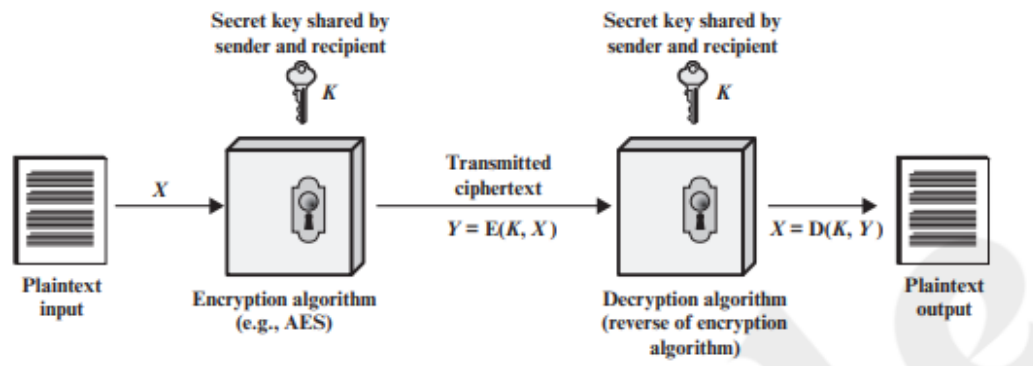
Question 8:

Explain the symmetric cipher model and its five elements. State the requirements for secure symmetric encryption.

Answer:

Symmetric Cipher Model

Symmetric encryption, also called **conventional encryption** or **single-key encryption**, is a method in which the **same secret key is shared by the sender and receiver** for encryption and decryption.



The basic model is:

1. Symmetric Cipher Model

Mathematically:

$$Y = E(K, X)$$

and

$$X = D(K, Y)$$

where:

- X = plaintext
- Y = ciphertext
- K = secret key
- E = encryption algorithm
- D = decryption algorithm

2. Five Elements of Symmetric Encryption

A symmetric encryption scheme has **five ingredients**.

1. Plaintext

Plaintext is the original, intelligible message or data that is given as input to the encryption algorithm.

It is represented by:

X

For example:

```
Plaintext = "HELLO"
```

2. Encryption Algorithm

The **encryption algorithm** performs various substitutions and transformations on the plaintext.

It takes:

Plaintext + Secret Key

and produces ciphertext:

$Y = E(K, X)$

The exact transformations performed depend on the key being used.

Examples of symmetric encryption algorithms include **AES** and **DES**.

3. Secret Key

The **secret key** is a value used by the encryption algorithm to control the transformations performed on the plaintext.

The key is:

- Independent of the plaintext
- Shared between sender and receiver
- Kept secret from the opponent
- Used in both encryption and decryption

For the same plaintext, changing the key produces a different ciphertext.

4. Ciphertext

The **ciphertext** is the scrambled message produced by the encryption algorithm.

It depends on both:

Plaintext + Secret Key

For a given plaintext, different keys produce different ciphertexts.

The ciphertext should appear as an unintelligible or apparently random stream of data to an unauthorized party.

5. Decryption Algorithm

The **decryption algorithm** is essentially the encryption algorithm operated in reverse.

It takes:

Ciphertext + Secret Key

and recovers the original plaintext:

$$X = D(K, Y)$$

The textbook explicitly defines decryption as the reverse of encryption, taking ciphertext and the secret key to produce the original plaintext.

3. Working of the Symmetric Cipher Model

Suppose the sender wants to send a message X.

Step 1: Plaintext

The sender starts with:

$$X = \text{Plaintext}$$

Step 2: Encryption

The sender applies the encryption algorithm using secret key K:

$$Y = E(K, X)$$

Step 3: Transmission

The resulting ciphertext Y is transmitted over the communication channel.

An opponent may obtain the ciphertext, but it should not be possible to recover the plaintext without the key.

Step 4: Decryption

The receiver applies the decryption algorithm using the same secret key:

$$X = D(K, Y)$$

Thus, the original plaintext is recovered.

4. Requirements for Secure Symmetric Encryption

There are **two requirements** for secure conventional/symmetric encryption.

Requirement 1: Strong Encryption Algorithm

The encryption algorithm must be sufficiently strong that an opponent who:

- Knows the encryption algorithm, and
- Has access to one or more ciphertexts

should not be able to:

- Decipher the ciphertext, or
- Discover the secret key.

The requirement is even stronger: the opponent should not be able to recover the key even if they possess **several ciphertexts together with their corresponding plaintexts**.

Therefore:

Strong algorithm + sufficiently large/ Secure key

is essential.

Requirement 2: Secure Key Distribution and Protection

The sender and receiver must obtain copies of the secret key through a **secure method** and must keep the key secure.

If an attacker obtains the key and knows the algorithm, then:

All communication using that key can be read

Therefore, in symmetric encryption, **key secrecy is the principal security problem.**

Question 9:

Explain the Caesar cipher, perform encryption using a specified shift and illustrate the brute-force attack.

Answer:

Caesar Cipher

1. Definition

The **Caesar cipher** is the earliest known and one of the simplest **substitution ciphers**. It replaces each plaintext letter with the letter located a fixed number of positions further down the alphabet. The alphabet is treated cyclically, so after **Z** comes **A**.

For example, with a shift of **3**:

```
Plain : ABCDEFGHIJKLMNOPQRSTUVWXYZ  
Cipher: DEFGHIJKLMNOPQRSTUVWXYZABC
```

Thus:

$A \rightarrow D, B \rightarrow E, C \rightarrow F$

and so on.

2. Mathematical Representation

Assign numerical values to the letters:

$$A=0, B=1, C=2, \dots, Z=25$$

For a shift of k , encryption is:

$$C = E(k, p) = (p + k) \bmod 26$$

where:

- p = numerical value of plaintext letter
- k = shift/key
- C = numerical value of ciphertext letter

The general Caesar encryption equation given in the text is $C = E(k, p) = (p + k) \bmod 26$

Decryption

The receiver subtracts the shift:

$$p = D(k, C) = (C - k) \bmod 26$$

3. Example of Encryption

Given:

Plaintext:

meet me after the toga party

Shift:

$$k=3$$

The substitution alphabet is:

Plain :	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Cipher:	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

For example:

$m \rightarrow P$

$e \rightarrow He$

$e \rightarrow He$

$t \rightarrow W$

Therefore:

meet $\rightarrow$ PHHW

The complete encryption is:

meet me after the toga party

↓

PHHW PH DIWHU WKH WRJD SDUWB

4. Illustration of Encryption

For the word **CAT**, using shift $k = 3$:

C

$$C = 2$$

$$(2 + 3) \bmod 26 = 5 = F$$

Therefore:

$$C \rightarrow F$$

A

$$A = 0$$

$$(0 + 3) \bmod 26 = 3 = D$$

Therefore:

$$A \rightarrow D$$

T

$$T = 19$$

$$(19 + 3) \bmod 26 = 22 = W$$

Therefore:

CAT → FDW

5. Brute-Force Attack on Caesar Cipher

A **brute-force attack** tries every possible key until the correct plaintext is found.

For the Caesar cipher, there are only:

25 possible keys

because a shift of 0 produces the original plaintext and the remaining useful shifts are 1 through 25. The Caesar cipher brute-force cryptanalysis can be performed by trying all 25 possible keys.

6. Example of Brute-Force Attack

Suppose the attacker intercepts:

PHHW PH DIWHU WKH WRJD SDUWB

The attacker does not know the shift.

They try all possible shifts:

```
Key 1 → OGGV OG CHVG VJG VQIC RCTVA
Key 2 → NFFU NF BGUU UIF UPHB QBSUZ
Key 3 → MEET ME AFTER THE TOGA PARTY
Key 4 → LDDS LD ZESDQ SGD SNFZ OZQSX
Key 5 → KCCR KC YDRCP RFC RMEY NYPRW
.
.
.
Key 25 → QIIX QI EJXIV XLI XSKE TEVXC
```

The correct plaintext is immediately recognizable at **Key 3**:

MEET ME AFTER THE TOGA PARTY

7. Why the Brute-Force Attack Works

Three characteristics make the attack easy:

1. Encryption and decryption algorithms are known

The attacker knows how the Caesar cipher works.

2. Very small keyspace

There are only:

25

possible keys.

Therefore, testing every key is trivial.

3. Plaintext language is recognizable

If the plaintext is normal English, the correct decryption can easily be identified because it produces meaningful words and sentences.

8. Why Caesar Cipher Is Not Secure

The Caesar cipher is **far from secure** because its keyspace is extremely small.

Even though the algorithm itself is simple, an attacker can simply test every possible shift:

1,2,3,...,25

and identify the meaningful plaintext.

Hence:

Small keyspace $\Rightarrow$ easy brute-force attack

9. Advantages and Limitations

Advantages

- Very simple to understand.

- Easy to implement.
- Requires minimal computation.
- Useful for illustrating the basic concept of substitution encryption.

Limitations

- Only **25 possible keys**.
 - Easily defeated by brute-force attack.
 - Plaintext is recognizable when the correct key is tried.
 - Provides essentially no security for modern communication.
-

Question 10:

Explain the principles of traditional block cipher structures, including Feistel structure, diffusion and confusion.

Answer:

Traditional Block Cipher Structures

A **block cipher** is a symmetric-key encryption technique in which plaintext is processed in **fixed-size blocks**. A key controls the transformation applied to each block to produce ciphertext.

Traditional block ciphers are based mainly on two important principles:

1. **Diffusion**
2. **Confusion**

A major structure used to achieve these principles is the **Feistel structure**.

1. Principles of Block Cipher Design

A good block cipher should make the relationship between the **plaintext**, **ciphertext**, and **key** as complex as possible.

Shannon identified two fundamental principles:

Diffusion + Confusion

These principles make statistical and cryptanalytic attacks difficult.

2. Diffusion

Diffusion means spreading the influence of each plaintext bit over many ciphertext bits.

The objective is to **hide the statistical structure of the plaintext** in the ciphertext.

For example, if one plaintext bit is changed, the change should eventually affect many ciphertext bits.

Thus:

Small change in plaintext → Large change in ciphertext

This property is closely related to the **avalanche effect**.

How diffusion is achieved

Diffusion is generally achieved using operations such as:

- Permutations
 - Bit transpositions
 - Shifts
 - Mixing operations
 - Multiple encryption rounds
-

3. Confusion

Confusion means making the relationship between the **key and ciphertext** as complex as possible.

The objective is to prevent an attacker from discovering the key by studying the relationship between plaintext and ciphertext.

A common mechanism for achieving confusion is the use of **substitution**, particularly nonlinear substitution.

Therefore:

Confusion → complex key - ciphertext relationship

Difference between diffusion and confusion

Diffusion	Confusion
Hides plaintext statistics	Hides relationship between key and ciphertext
Spreads plaintext influence	Makes key relationship complex
Often achieved using permutations/mixing	Often achieved using substitutions/S-boxes
Affects many ciphertext bits from plaintext changes	Makes key recovery difficult

4. Feistel Structure

The **Feistel structure** is a widely used design structure for block ciphers.

A block of plaintext is divided into two equal halves:

L_0, R_0

The two halves are processed through a number of **rounds**.

In each round:

- One half is passed through a round function.
- The result is combined with the other half using XOR.
- The two halves are then exchanged.

5. Feistel Round

A typical Feistel round can be represented as:

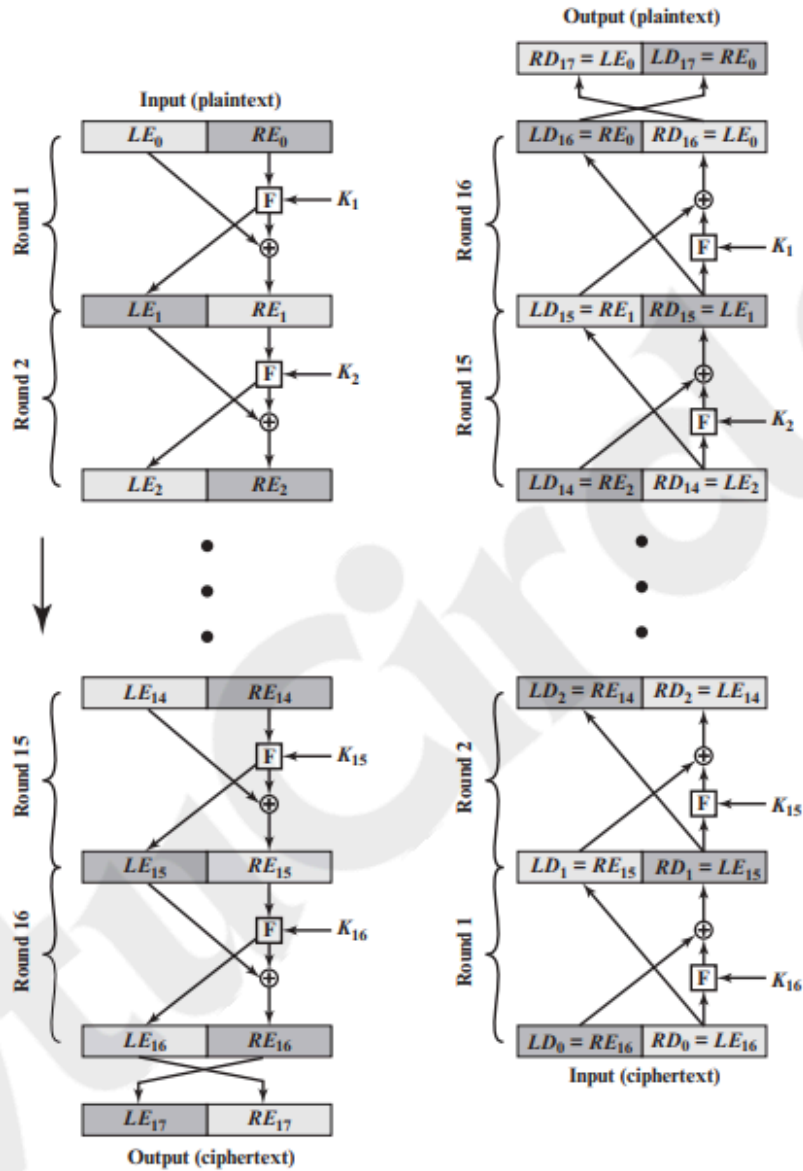


Figure 4.3 Feistel Encryption and Decryption (16 rounds)

More precisely, for round i :

$$L_i = R_{i-1}$$

and

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

where:

- L_i = left half after round i
- R_i = right half after round i

- K_i = round key
- F = round function
- $\oplus$ = XOR

6. Working of a Feistel Structure

Suppose the plaintext block is divided into:

$$P = (L_0, R_0)$$

For the first round:

$$L_1 = R_0$$

$$R_1 = L_0 \oplus F(R_0, K_1)$$

For the second round:

$$L_2 = R_1$$

$$R_2 = L_1 \oplus F(R_1, K_2)$$

The process continues for several rounds.

After the final round, the two halves are combined to produce the ciphertext.

7. Major Advantage of Feistel Structure

One of the most important features of the Feistel structure is that **the encryption and decryption processes use essentially the same structure.**

The main difference is the **order in which the round keys are used.**

For encryption:

$$K_1, K_2, K_3, \dots, K_n$$

For decryption:

$$K_n, K_{n-1}, K_{n-2}, \dots, K_1$$

This simplifies the implementation of both encryption and decryption.

8. Avalanche Effect

A desirable property of a block cipher is the **avalanche effect**.

It means that a small change in the plaintext or key should produce a significant change in the ciphertext.

For example:

```
Plaintext 1: 10110010
Plaintext 2: 10110011
                ↑
            One bit changed
```

After encryption:

```
Ciphertext 1: 01101001
Ciphertext 2: 11010110
```

Many ciphertext bits may change even though only one plaintext bit was changed.

This demonstrates effective **diffusion**.

9. Summary

Concept	Meaning	Main Purpose
Block cipher	Encrypts fixed-size blocks	Secure data encryption
Feistel structure	Splits block into two halves and processes them through rounds	Provides a practical cipher structure
Diffusion	Spreads plaintext influence over many ciphertext bits	Hides plaintext statistics
Confusion	Complicates relationship between key and ciphertext	Makes key recovery difficult
Avalanche effect	Small input change causes large output change	Indicates strong diffusion

